

# 第十章复习 - 答案.

(1) C.

比式和根式判例 - 一定是极限值小于1

$$\sqrt[n]{|a_n|} = (1 - \frac{a}{n})^n \rightarrow e^{-a} < 1 \text{ if } a > 0. \quad \textcircled{2} (1 + \frac{1}{n})^n \rightarrow e, (1 + \frac{x}{n})^n \rightarrow e^x, n \rightarrow \infty$$

(2) A.

$$A: \frac{a_n}{(1+\ln n)^2} < \frac{1}{n(1+\ln n)^2}, \quad \int_1^{\infty} \frac{dx}{x(1+\ln x)^2} = -\frac{1}{1+\ln x} \Big|_1^{\infty} = \frac{1}{2}$$

$$\Rightarrow \sum_{n=1}^{\infty} (-1)^n \frac{a_n}{(1+\ln n)^2} \text{ 绝对收敛.}$$

$$0 < a_n \leq \frac{1}{n} \Rightarrow a_n > 0$$

$$a_n \rightarrow 0$$

但  $a_n$  不满足

收敛

$$B: \text{let } a_n = \begin{cases} \frac{1}{2n}, & n=2k \\ 0, & n=2k+1 \end{cases}$$

$$0, n=2k+1$$

$$C: a_n = \frac{1}{2n}$$

$$D: a_n = \frac{1}{2n} \quad \frac{1}{2} \int_1^{\infty} \frac{dx}{x(1+\ln x)} = \frac{1}{2} [\ln|1+\ln x|]_1^{\infty} = \infty$$

(3) C.

在收敛区间 (开区间) 内可逐项求导, 逐项积分

$$\sum a_n x^n \text{ 在 } x=2 \text{ 收敛} \Rightarrow \sum a_n x^n, (\sum a_n x^n)' = \sum n a_n x^{n-1} \text{ 在 } |x| < 2 \text{ 收敛}$$

$$\Rightarrow \sum n a_n x^{n-1} \text{ 在 } x=1 \text{ 收敛}$$

端点要单独看

(4) D.

$$\sum a_n \text{ 收敛}, \sum a_{n+1} \text{ 收敛} \Rightarrow \sum \frac{a_n + a_{n+1}}{2} = \frac{1}{2} \sum (a_n + a_{n+1}) \text{ 收敛}$$

(5) C.

$$a_n \rightarrow 0 \Rightarrow a_n \text{ 有界}, |a_n| \leq M$$

比较判例法要求  $a_n > 0$ , 排除 A, B.

$$\text{If } \sum |b_n| \text{ 收敛} \Rightarrow b_n \rightarrow 0 \Rightarrow |b_n| < 1, n > N \Rightarrow a_n^2 b_n^2 < M^2 |b_n| \Rightarrow \sum a_n b_n \text{ 收敛}$$

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在收敛区间 (开区间) 内可逐项求导, 求积分, 开区间

$$\sum a_n x^n \text{ 在 } x=2 \text{ 收敛} \Rightarrow \sum a_n x^n, (\sum a_n x^n) = \sum n a_n x^{n-1} \text{ 在 } |x| < 2 \text{ 收敛}$$

$$\Rightarrow \sum n a_n x^{n-1} \text{ 在 } x=1 \text{ 收敛.}$$

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(6) C

$$na_n = \frac{a_n}{n} \rightarrow 1 > 0 \Rightarrow \sum a_n \text{ 发散.}$$

(7) C

$$\left| \frac{\sin(n+a)}{n^{1.01}} \right| \leq \frac{1}{n^{1.01}} \text{ 绝对收敛.}$$

 $\Rightarrow$  收敛.

$$\left[ \sum_{x=2}^{\infty} \frac{dx}{x \ln x} = \ln(\ln x) \right]_2^{\infty} = \infty \text{ 发散}$$

(8) D.

$$\sqrt[n]{|u_n|} \rightarrow \frac{|x|}{3} < 1 \Rightarrow -3 < x < 3 \Rightarrow [-3, 3)$$

$$x=3 \sum \frac{1}{n} \text{ 发散}; x=-3, \sum \frac{(-1)^n}{n} \text{ 条件收敛}$$

(9) B

$$\left| \frac{\sin(n)}{n^2} \right| \leq \frac{1}{n^2} \text{ 收敛}, \sum \frac{(-1)^n}{n+1} \text{ 条件收敛} \Rightarrow \text{收敛. but } \left| \frac{\frac{\sin(n)}{n^2} + \frac{(-1)^n}{n+1}}{n} \right| =$$

$$\sum \frac{(-1)^n}{n+1} \text{ 条件收敛.}$$

不绝对收敛.

(10) B.

$$\frac{\frac{1}{n^{1+\frac{1}{n}}}}{\frac{1}{n}} = n^{\frac{1}{n}} \rightarrow 1, \sum \frac{1}{n} \text{ 发} \Rightarrow \sum \frac{1}{n^{1+\frac{1}{n}}} \text{ 发}$$

$$A. \sum \frac{n^2}{2^n}, \sqrt[n]{a_n} = \frac{\sqrt[n]{n^2}}{2} \rightarrow \frac{1}{2} < 1 \text{ 收}$$

$$C. \begin{cases} \frac{\ln n}{\sqrt{n}} > 0, \rightarrow 0, \downarrow \\ \frac{\ln n}{\sqrt{n}} > \frac{1}{\sqrt{n}} \text{ 发} \end{cases} \Rightarrow \text{条件收}$$

(11) D.

$$0 < a_n^2 < \frac{1}{n^2}, \sum a_n^2 \text{ 收敛}$$

$$0 < a_n < \frac{1}{n} \rightarrow a_n > 0 \text{ but } a_n \text{ 单调性不确定.}$$

$$D. \left| \frac{(3+(-1)^n 2)^n}{6^n} \right| < \frac{5^n}{6^n} = \left( \frac{5}{6} \right)^n \Rightarrow \text{绝对收敛.}$$

2. (1)  $\lim_{n \rightarrow \infty} \left( \tan \frac{1}{n} - k \ln \left( 1 + \frac{1}{n} \right) \right) = 0 \Rightarrow \left( \frac{1}{n} + o\left(\frac{1}{n}\right) \right) - k \left( \frac{1}{n} + o\left(\frac{1}{n}\right) \right) = (k+1) \frac{1}{n} + o\left(\frac{1}{n}\right)$   
 ~~$\Rightarrow k+1=0 \Rightarrow k=-1$~~

(2)  $\lim_{x \rightarrow 0} \frac{3 \ln(2x) - 2 \ln(3x)}{6x^3 + x^4} = \lim_{x \rightarrow 0} \frac{3(2x - \frac{(2x)^3}{3!} + \dots) - 2(3x - \frac{(3x)^3}{3!} + \dots)}{6x^3 + x^4}$   

$$= \frac{-3 \cdot \frac{8}{6} + 2 \cdot \frac{27}{6}}{6} = \frac{-4 + 9}{6} = \frac{5}{6}$$

(3)  $\lim_{x \rightarrow 0} \frac{\frac{\sin x}{x} - x}{(\cos x - 1)(e^{2x} - \cos x)} = \lim_{x \rightarrow 0} \frac{(x - \frac{x^3}{3!} + \dots) - x}{\left[1 - \frac{x^2}{2!} + \dots\right] \left[1 + 2x + \frac{(2x)^2}{2!} + \dots - (x - \frac{x^3}{3!} + \dots)\right]} \rightarrow \frac{-\frac{1}{6}}{-\frac{1}{6}} = \frac{1}{6}$

(4)  $\sum_{n=1}^{\infty} \frac{1}{2^n \cdot n!} = \sum_{n=1}^{\infty} \frac{1}{n!} \left(\frac{1}{2}\right)^n = \sum_{n=1}^{\infty} \frac{x^n}{n!}, x = \frac{1}{2}$   

$$= e^x - 1 = e^{\frac{1}{2}} - 1$$

(5)  $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n!} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!}, x = -1$   

$$= -\frac{1}{2} = e^x - 1 = e^{-1} - 1 = -\frac{1}{2}$$

(6)  $\sum_{n=1}^{\infty} \frac{x^{2n+2}}{n(n+4)2^n}$  let  $2n+2=6 \Rightarrow n=2$   $\frac{1}{2 \times 6 \cdot 2^2} x^6$   

$$\Rightarrow \frac{f^{(6)}(0)}{6!} = a_6 = \frac{1}{48} \Rightarrow f^{(6)}(0) = \frac{1 \times 2 \times 3 \times 4 \times 5 \times 6}{48} = 15$$

3. (1) T.  $\sum a_n$  收敛,  $a_n < 0, \Rightarrow \sum |a_n|$  收敛,  $a_n \rightarrow 0, \Rightarrow a_n^2 \leq |a_n|$

(2) T. 在  $x=0$  收敛  $\Rightarrow$  绝对收敛  $|x-1| < |0-1| = 1 \Rightarrow 0 < x < 2$   
 $x=2$  发散:  $|x-1| > |2-1| \Rightarrow x > 2$  或  $x < 0$